

Numpy



* Numpy

Numpy stands for Numerical python. It is a python library mainly used for numerical calculations and working with arrays. It provides useful functions for performing mathematical, statistical operations easily and efficiently.

eg:-

```
import numpy as np
a = np.array([10, 20, 30])
print(a)
```

output

[10 20 30]

* Numpy Array

A Numpy array is a collection of elements stored in an organized structure. Unlike a normal python list, a Numpy is designed especially for numerical calculations and can perform operations on multiple elements at the same time.

eg:-

```
import numpy as np
a = np.array([10, 20, 30, 40])
print(a)
```

output

[10 20 30 40]

* `np.arange()`

`np.arange` is a numpy function used to create an array containing numbers within a specified range. we can also provide a step value to control the difference between consecutive numbers.

eg:-

```
import numpy as np
```

```
a = np.arange(1, 10, 2)
```

```
print(a)
```

output

```
[1 3 5 7 9]
```

* `reshape()`

The `reshape()` function is used to change the shape or dimensions of an array without changing data.

eg import numpy as np

```
a = np.array([2, 2, 3, 4, 5, 6])
```

```
b = a.reshape(2, 3)
```

```
print(b)
```

output

```
[[2 2 3]
```

```
[4 5 6]]
```

* `np.ones()`

`np.ones` is a numpy function used to create an array in which all the elements are 1. It is useful when we need an array or matrix initialized with ones.

eg:-

```
import numpy as np
a = np.ones((2,3))
print(a)
```

output

```
[[1. 1. 1.]
 [1. 1. 1.]]
```

* `np.zeros()`

`np.zeros` is used to create an array in which all the elements are 0. It is commonly used to initialize an array before storing values in it.

* `np.linspace()`

`np.linspace` is used to create an array containing a specified number of evenly spaced values between a starting value and ending value.

eg:-

```
import numpy as np
a = np.linspace(0, 10, 5)
print(a)
```

```
[0. 2.5 5. 7.5 10.]
```

* `np.identity()`

`np.identity()` is used to create a square identity matrix. In an identity matrix, all diagonal elements are 1 and all other elements are 0.

* Indexing

Indexing is the process of accessing a particular element from an array using its position. In numpy, indexing starts from 0, so the first element has index 0, the second has index 1, and so on.

eg:-

```
import numpy as np
a = np.array([10, 20, 30, 40])
print(a[0])
print(a[2])
```

output

10

30

* Slicing

Slicing means taking a part of an array.

`array[start:stop:step]`

start → where to start

stop → where to stop

step → how many positions to jump

eg:-

```
a = np.array([10, 20, 30, 40, 50])
print(a[1:4])
```

output

`[20, 30, 40]` → Because index 1, 2, 3 are selected

using step

```
a = np.array([10, 20, 30, 40])
print(a[0:5:2])
```

output `[10, 30, 50]`

Here 2 means jump by 2 positions.

* $axis$.

An $axis$ represents the direction in which an operation is performed on an array. In a 2D array, $axis=0$ perform the operation column-wise, while $axis=1$ perform the operation row-wise.

* Transpose

Transpose is an operation that changes the rows of an array into columns and the columns into rows. It is commonly used when working with matrices.